Comp 512
Spring 2012

A Deeper Look at Data-Flow Analysis

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Data-flow Analysis

Definition

Data-flow analysis is a collection of techniques for *compile-time* reasoning about the *run-time* flow of values

- We use the results of DFA to prove safety & identify opportunities
  - → Not an end unto itself
- Almost always involves building a graph
  - → Control-flow graph, call graph, or derivatives thereof
  - → Sparse evaluation graphs to model flow of values *(efficiency)*
- Usually formulated as a set of *simultaneous equations*
  - → Sets attached to nodes and edges
  - → Often use sets with a lattice or semilattice structure
- Desired result is usually *meet over all paths* solution
  - → "What is true on every path from the entry?"
  - → "Can this happen on any path from the entry?"

Data-flow Analysis

We have seen two data-flow problems: Dom and Live

Computing Dominators

- Domain is nodes in the flow graph being analyzed
- Simple set of data-flow equations

N is the set of nodes in the flow graph

Initialization

$$DOM(n_0) = \{ n_0 \}$$

$$DOM(n) = N, \forall n \neq n_0$$

Fixed-point equation

$$DOM(n) = \{ n \} \cup (\cap_{p \in preds(n)} DOM(p))$$

Can solve equations solve them with any data-flow solver

Data-flow Analysis

Computing Live variables

- Domain is the set of variable names in the procedure
- Data-flow equations are more complex

Initialization

$$LIVEOUT(n) = \emptyset, \ \forall n$$

Fixed-point equations

$$LIVEOUT(\mathrm{b}) = \bigcup_{s \in succ(b)} LIVEIN(\mathrm{s})$$

$$LIVEIN(b) = UEVAR(b) \cup (LIVEOUT(b) \cap \overline{VARKILL(b)})$$

where

UEVAR(b) is the set of names used in b before definition in b

VARKILL(b) is the set of names defined in b

Classic Algorithm: Round-robin Iterative Algorithm

$DOM(n_0) \leftarrow \{ n_0 \}$

for i $\leftarrow$ 1 to $|N|$

  $DOM(n_i) \leftarrow \{ N \}$

change $\leftarrow$ true

while (change)

  change $\leftarrow$ false

  for i $\leftarrow$ 0 to $|N|$

    $TEMP \leftarrow \{ n_i \} \cup (\cap_{p \in pred(ni)} DOM(p))$

if $DOM(n_i) \neq TEMP$ then

    change $\leftarrow$ true

    $DOM(n_i) \leftarrow TEMP$

Just the fixed-point equation

Very Simple Algorithm

- Halts when *DOM* sets stop changing
- Makes successive sweeps over the nodes in some fixed order

  $\rightarrow$ i $\leftarrow$ 0 to $|N|$

Solving a Data-flow Problem

To compute Dominator sets

- We need to build the control-flow graph
  - → Defines *predecessors* and *successors*
- Run the round-robin worklist algorithm
  - → Initializes *DOM(n)* for each node *n*
  - → Iterates until it reaches a fixed point (*e.g.*, *DOM* stabilizes)

To solve another data-flow problem

- Replace the initialization step and the fixed-point equation
- Fixed-point equation includes direction of propagation
  - → *Predecessors* or *successors*, as needed

To explain data-flow analysis, Kildall introduced a lattice-theoretic model. Kam & Ullman (among others) developed specific formulations for *iterative* data-flow algorithms

See J.B. Kam and J.S. Ullman, "Global Data Flow Analysis and Iterative Algorithms", *JACM* 23(1), January 1976, pp. 158-171

Classic Algorithm: Round-robin Iterative Algorithm

$DOM(n_0) \leftarrow \emptyset$

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if $DOM(n_i) \neq TEMP$ then

    change $\leftarrow$ true

    $DOM(n_i) \leftarrow TEMP$

Just the fixed-point equation

Questions we must ask

- Termination: does it halt?
- Correctness: what answer does it produce?
- Speed: how quickly does it find that answer?

Data-flow Analysis

The basics

- Data-flow sets are drawn from a semi-lattice, L, of facts
- Sets are modified by transfer functions, f_i, that model effect of code on contents of the sets
  - → Function space of all possible transfer functions is F
- Properties of L and F govern termination, correctness, & speed

To reason about the properties of a (*proposed*) data-flow problem, we cast it into a lattice-theory framework and prove some simple theorems about the problem.

Data-flow Analysis

Limitations

1. Precision – *"up to symbolic execution"*
   → Assume all paths are taken
2. Solution – cannot afford to compute MOP solution
   → Large class of problems where MOP = MFP= LFP
   → Not all problems of interest are in this class
3. Arrays – treated naively in classical analysis
   → Represent whole array with a single fact
4. Pointers – difficult (*and expensive*) to analyze
   → Imprecision rapidly adds up
   → Need to ask the right questions

Good news:

Simple problems can carry us pretty far

Summary

For scalar values, we can quickly solve simple problems

*

Data-flow Analysis

Semilattice

A semilattice $L = (S, \wedge)$ is a set S and a meet operation $\wedge$ such that,

$\forall\ a, b, \&\ c \in L$:

1. $a \wedge a = a$
2. $a \wedge b = b \wedge a$
3. $a \wedge (b \wedge c) = (a \wedge b) \wedge c$

The meet operator combines the sets when two paths converge, or meet.

$\wedge$ imposes a partial order on L, $\forall\ a, b, \&\ c \in L$:

1. $a \geq b \Leftrightarrow a \wedge b = b$
2. $a > b \Leftrightarrow a \geq b$ and $a \neq b$

A semilattice has a bottom element, denoted $\perp$

1. $\forall\ a \in L, \perp \wedge a = \perp$
2. $\forall\ a \in L$, a $\geq \perp$

Sometimes we work with a lattice, which has a top element, denoted

$\forall\ a \in L, \top \wedge a = a$

and another operator, called join $\vee$

Data-flow Analysis

How does this relate to data-flow analysis?

- Choose a semilattice to represent the facts
- Attach a meaning to each $a \in L$

  Each $a \in L$ is a distinct set of known facts
- With each node n, associate a function $f_n : L \rightarrow L$

  f_n models behavior of code in block corresponding to n
- Let F be the set of all functions that the code might generate

2^N is the powerset of N, the set of all subsets of N

Example — DOM

- Semilattice is $(2^N, \wedge)$, where N is the set of nodes in the flow graph and $\wedge$ is $\cap$, and $\perp$ is $\emptyset$
- For a node n, f_n has the form $f_n(x) = x$

World's simplest data-flow equation

Data-flow Analysis

How does this relate to data-flow analysis?

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- Attach a meaning to each $a \in L$

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Example — Live

- Semilattice is $(2^{Vars}, \wedge)$, where *Vars* is the set of names in the code and $\wedge$ is $\cup$, and $\perp$ is *Vars*
- For a node n, f_n has the form $f_n(x) = a \cup (x \cap b)$, where a & b are constants (*UEVAR & VARKILL respectively*)

A particularly common form for a data-flow equation

Iterative Data-flow Analysis

Termination

- If every $f_n \in F$ is monotone, *i.e.*, , $f(x \wedge y) \leq f(x) \wedge f(y)$, <u>and</u>
- If the lattice is <u>bounded</u>, *i.e.*, every descending chain is finite
  - Chain is sequence $x_1, x_2, \ldots, x_n$ where $x_i \in L$, $1 \leq i \leq n$
  - $x_i > x_{i+1}$, $1 \leq i < n \Rightarrow$ chain is <u>descending</u>

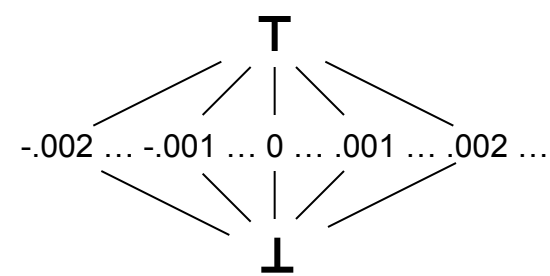
Finite lattice, bounded descending chains, & monotone functions $\Rightarrow$ termination

Both DOM & LIVE have monotone transfer functions & finite (bounded) semilattices.

Then

- The set at each node can only change a finite number of times
- The iterative algorithm must halt on an instance of the problem ♦

- Any finite semilattice is bounded
- Some infinite semilattices are bounded



Real constants

Iterative Data-flow Analysis

Correctness

(What does it compute?)

- If every $f_n \in F$ is monotone, *i.e.*, $f(x \wedge y) \leq f(x) \wedge f(y)$, <u>and</u>
- If the semilattice is <u>bounded</u>, *i.e.*, every descending chain is finite
  - Chain is sequence $x_1, x_2, \ldots, x_n$ where $x_i \in L$, $1 \leq i \leq n$
  - $x_i > x_{i+1}$, $1 \leq i < n \Rightarrow$ chain is <u>descending</u>

$f^k(x)$ is the application of f to x k times

Given a bounded semilattice S and a monotone function space F

- $\exists\, k$ such that $f^k(\bot) = f^j(\bot)\ \forall\, j > k$
- $f^k(\bot)$ is called the least fixed-point of f over S
- If L has a $\top$, then $\exists\, k$ such that $f^k(\top) = f^j(\top)\ \forall\, j > k$ and $f^k(\top)$ is called the maximal fixed-point (MFP) of f over S

optimism

MFP is largest fixed point, defined by ≥; LFP is smallest fixed point.

Iterative Data-flow Analysis

Correctness

- If every $f_n \in F$ is monotone, *i.e.*, $f(x \wedge y) \leq f(x) \wedge f(y)$, <u>and</u>
- If the lattice is <u>bounded</u>, *i.e.*, every descending chain is finite
  - → Chain is sequence $x_1, x_2, \ldots, x_n$ where $x_i \in L$, $1 \leq i \leq n$
  - → $x_i > x_{i+1}$, $1 \leq i < n$ ⇒ chain is <u>descending</u>

Then

- The round-robin algorithm computes a least fixed-point (LFP)
- The uniqueness of the solution depends on other properties of *F*

- Unique solution ⇒ it finds the one we want
- Multiple solutions ⇒ we need to know which one it finds

Iterative Data-flow Analysis

Correctness

- Does the iterative algorithm compute the desired answer?

Admissible Function Spaces

1. $\forall f \in F, \forall x,y \in L, f(x \wedge y) = f(x) \wedge f(y)$
2. $\exists f_i \in F$ such that $\forall x \in L, f_i(x) = x$
3. $f,g \in F \; \exists h \in F$ such that $h(x) = f(g(x))$
4. $\forall x \in L, \exists$ a finite subset $H \subseteq F$ such that $x = \wedge_{f \in H} f(\perp)$

If meet does not distribute over function application, then the fixed point solution *may not be* unique. The iterative algorithm will find a LFP.

If F meets these four conditions, then an instance of the problem will have a unique fixed point solution *(instance $\Rightarrow$ graph + initial values)*

$\Rightarrow$ LFP = MFP = MOP

$\Rightarrow$ order of evaluation does not matter

Both DOM & LIVE meet all four criteria

*

Iterative Data-flow Analysis

If a data-flow framework meets those admissibility conditions then it has a unique fixed-point solution

- The iterative algorithm finds the (best) answer
- The solution does not depend on order of computation
- Algorithm can choose an order that converges quickly

Intuition

- Choose an order that propagates changes as far as possible on each "sweep"
  - → Process a node's predecessors before the node
- Cycles pose problems, of course
  - → Ignore back edges when computing the order?

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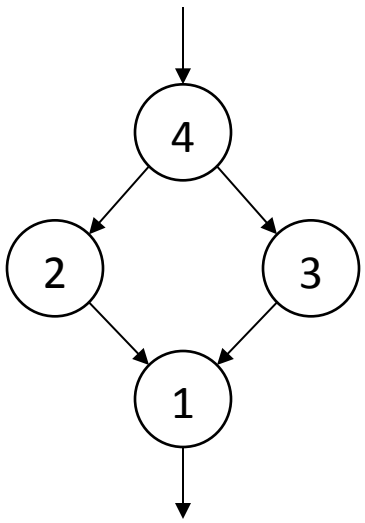
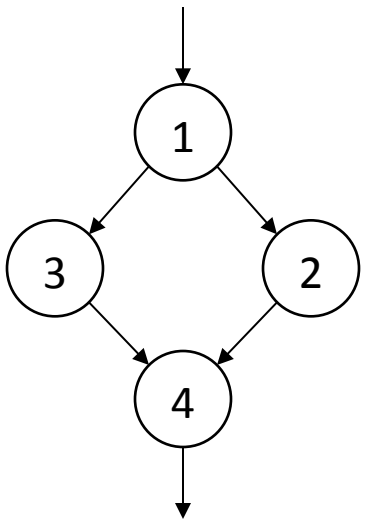
Ordering the Nodes to Maximize Propagation



Postorder



Reverse Postorder

N+1 - *postorder number*

- Reverse postorder visits predecessors before visiting a node
- For backward problems, compute reverse postorder on the reverse CFG
  - → Reverse preorder is not necessarily the same

See Exercise 9.4 in EaC2e

Data-flow Analysis

Complexity

- For a problem with an admissible function space & a bounded semilattice,
- If the functions all meet the *rapid condition*, *i.e.*,

$$\forall f,g \in F, \forall x \in L, f(g(\perp)) \geq g(\perp) \wedge f(x) \wedge x$$

Sets stabilize in two passes around a cycle.

then, a round-robin, reverse-postorder iterative algorithm will halt in d(G)+3 passes over a graph G

Each "pass" does **O**(*E*) meets and **O**(*N*) other operations

d(G) is the loop-connectedness of the graph w.r.t a *DFST*

→ Maximal number of back edges in an acyclic path

→ Several studies suggest that, in practice, d(G) is small (*<3*)

→ For most CFGs, d(G) is independent of the specific *DFST*

Kam and Ullman, JACM, 1976. The paper gives a couple of equivalent formulations of the *rapid condition.*

Data-flow analysis

What does this mean?

- Reverse postorder
  - → Number the nodes in a postorder traversal
  - → Reverse the order
- Round-robin iterative algorithm
  - → Visit all the nodes in a consistent order (RPO)
  - → Do it again until the sets stop changing

So, these conditions are easily met

- → Admissible framework, rapid function space
- → Round-robin, reverse-postorder, iterative algorithm

⇒The analysis runs in (*effectively*) linear time

Data-flow Analysis

How do we use these results?

- Prove that data-flow framework is admissible & rapid
  - → Its just algebra
  - → Most (but not all) global data-flow problems are rapid
  - → This is a property of *F*

- Code up the iterative algorithm
  - → World's simplest data-flow algorithm
  - → Other versions (worklist) have similar behavior

This lets us ignore most of the other data-flow algorithms in 512

DOM, LIVE, and most global data-flow problems are *admissible* & *rapid*

Comp 512 Lab

Time to Make Decisions

Options:

- Each team does the same basic study with a different compiler
  - → Standardize on language(s), benchmarks, and tests
  - → Vary the compilers
- Each team does a similar, but different, study
  - → Standardize on methodology
  - → Teams choose a language, compiler, and benchmarks

Either option should lead to an interesting learning experience

Data-flow Analysis

How do we use these results?

- Prove that data-flow framework is admissible & rapid
  - → Its just algebra
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How about some examples of problems that are *not* rapid?

Flow-Insensitive Interprocedural Summary Problems

MOD: $v \in$ MOD(s) $\Leftrightarrow$ executing s might change v's value

REF: $v \in$ REF(s) $\Leftrightarrow$ executing s might change v's value

These formulations can be solved with a flow-insensitive method

The equations

$$\text{GMOD}(p) = \text{IMOD}(p) \cup \cup_{e = \langle p,q \rangle} b_e(\text{GMOD}(q))$$

$$\text{DMOD}(e) = b_e(\text{GMOD}(q))$$

Where

GMOD(p): variables that may be modified by an execution of p

IMOD(p): variables that may be modified locally in p

DMOD(s): variables that may be modified by call site s, ignoring aliasing

b_e: a function that inverts the formal parameter $\rightarrow$ actual parameter binding

These functions do not have the "classic" form:

$$a = b \cup (c \cap d)$$

b_e makes the problem *non-rapid* on the call graph.

Solve on either the call graph or the binding graph β

Cooper & Kennedy, PLDI 88, June, 1988

Flow-Insensitive Interprocedural Summary Problems

MOD: $v \in$ MOD(s) $\Leftrightarrow$ executing s might change v's value

REF: $v \in$ REF(s) $\Leftrightarrow$ executing s might change v's value

Worst case examples

```
fee : procedure (a, b, c, d, e);
        integer a, b, c, d, e, f;
        …
        f = 0;
        ….
        call fee(b, c, d, e, f);
        end;
```

The "shift register"

Flow of values through the procedure

Interprocedural flow from binding at the call site

Flow-Insensitive Interprocedural Summary Problems

MOD: $v \in$ MOD(s) $\Leftrightarrow$ executing s might change v's value

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Worst case examples

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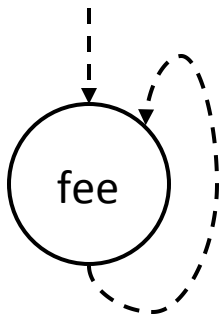
The "shift register"



Call graph (fragment)

How many iterations does it take to determine that the value of a can change due to a call to fee()?

The answer depends on the number of edges in the cycle of bindings — which is unrelated to the call graph.

(Add another 5 parameters)

Flow-Insensitive Interprocedural Summary Problems

To solve these problems quickly:

1. Build a graph, β, that reflects the binding patterns of the parameters
   a. Node for each formal parameter and each actual parameter
   b. Edges for each binding at each call site
2. MOD & REF for parameters are now reachability on β
3. Solve the problem for global variables with a simple data flow equations

$$\text{GMOD}_G(p) = \text{IMOD}_G(p) \cup_{e = (p,q) \in CG} \text{GMOD}_G(q)$$

$$\text{DMOD}_G(e=(p,q)) = \text{GMOD}_G(q)$$

Steps 1 & 2 are linear in the size* of β.

Step 3 takes d(CG) + 3 passes, in an RPO traversal of the RCFG

Arguably, linear time or near linear time

* β is linearly larger than the call graph, as long as MaxParms is bounded.

Example

|CONSTANTS| = |*variables*|

Transformation: Global Constant Folding

- Along every path to *p*, *v* has same known value
- Specialize computation at *p* based on *v*'s value

Data-flow problem: Constant Propagation

Domain is the set of pairs $<v_i,c_i>$ where v_i is a variable and $c_i \in C$

$$\text{CONSTANTS}(b) = \wedge_{p \in preds(b)} f_p(\text{CONSTANTS}(p))$$

- $\wedge$ performs a pairwise meet on two sets of pairs
- $f_p(x)$ is a block specific function that models the effects of block p on the $<v_i,c_i>$ pairs in *x*

Form of *f* is quite different than in *LIVE*

Constant propagation is a forward flow problem

Classic formulation before Reif-Lewis

Example

Meet operation requires more explanation

- $c_1 \wedge c_2 = c_1$ if $c_1 = c_2$, else $\perp$ (*bottom & top as expected*)

What about f_p ?

- If p has one statement then

  $x \leftarrow y$ with CONSTANTS(p) = {…<x,i_1>,…<y,i_2>…}

  then f_p(CONSTANTS(p)) = CONSTANTS(p) - <x,i_1> + <x,i_2>

  $x \leftarrow y$ *op* z with CONSTANTS(p) = {…<x,i_1>,…<y,i_2>,…<z,i_3>…}

  then f_p(CONSTANTS(p)) = CONSTANTS(p) - <x,i_1> + <x,i_2 *op* i_3>

- If p has n statements then

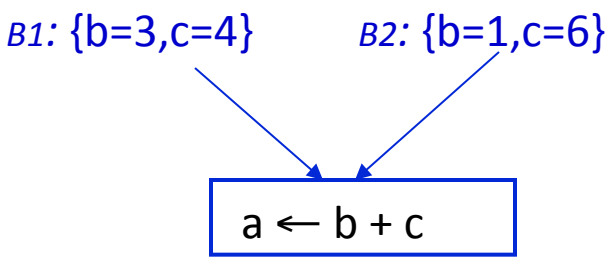
  $f_p(\text{CONSTANTS}(p)) = f_n(f_{n-1}(f_{n-2}(\ldots f_2(f_1(\text{CONSTANTS}(p)))\ldots)))$

  where f_i is the function generated by the ith statement in p

f_p interprets p over CONSTANTS

Some problems are not admissible

Global constant propagation



- Function "f" models block's effects
- f(*B1*) = {a=7,b=3,c=4}
- f(*B2*) = {a=7,b=1,c=6}
- f(*B1*∧*B2*) = Ø

- First condition in admissibility

  $\forall f \in F, \forall x,y \in L, f(x \wedge y) = f(x) \wedge f(y)$
- Constant propagation is not admissible
  - → Kam & Ullman time bound does not hold
  - → There are tight time bounds, however, based on lattice height
  - → Require a variable-by-variable formulation …
  - → The problem has multiple fixed-point solutions!

Comp 512 Lab

Some Suggested Benchmarks

Name	*Language*	*Codes*	*SLOCs*	*Source*
NoFib	Haskell	~30	8 to 1,500	Glasgow
OOPack	C++	4	Kernels	LBL
CSIBE	C	19	?	U Szeged
MiBench	C	38		U Mich
Polyhedron	Fortran 90	17	?	Polyhedron Labs
MediaBench I	C	11		Consortium
DaCapo	Java	14		DaCapo Project
Caffeine	Java	9		Pendragon Software
Chrome	C/C++	1	Lots	Google
Individual Codes	C?	Avida (Mich ST) Owl (RICE)		

Will Partain

Comp 512 Lab

Some Suggested Compilers

- gcc
- Llvm
- Open64
- IBM XL C compiler for POWER7
- Safe C compiler for web applications (google project?)
- GHC — the Glasgow Haskell Compiler